

Effects of Electric Current

I] Textbook Questions:

★ 1 Mark Objectives: ★

1. Tell the odd one out.

- a. Fuse wire, bad conductor, rubber gloves, generator.
- b. Voltmeter, Ammeter, galvanometer, thermometer.
- c. Loud speaker, microphone, electric motor, magnet.

Ans: Generator [2025]

Ans: Thermometer

Ans: Magnet [2023, 2025]

2. Electromagnetic induction means

- a. Charging of an electric conductor.
- b. Production of magnetic field due to a current flowing through a coil.
- c. **Generation of a current in a coil due to relative motion between the coil and the magnet. (Ans)**
- d. Motion of the coil around the axle in an electric motor.

★ 2 Marks Questions: ★

1. Give Scientific reasons.

a. Tungsten metal is used to make a solenoid type coil in an electric bulb. [2023, 2024, 2025]

Ans:

1. Equipment such as electric bulb makes use of the heating effect of electric current.
2. Electrical conductors having higher resistivity are used here.
3. A tungsten wire is used to make a solenoid type coil in an electric bulb for this purpose.
4. Because of the current, this wire gets heated (to nearly 3400°C) and emits light. The hot wire also radiates heat to a certain extent.

b. In the electric equipment producing heat e.g. iron, electric heater, boiler, toaster etc. an alloy such as Nichrome is used, not pure metals.

Ans:

1. Equipment such as iron, electric heater, boiler, toaster etc make use of the heating effect of electric current.
2. Electrical conductors having higher resistivity are used here.
3. A coil made up of an alloy Nichrome is used for this purpose. Nichrome offers high resistance than pure metals
4. Therefore, it produces high amount of heat when current is passed through it.
Hence, In the electric equipment producing heat e.g. iron, electric heater, boiler, toaster etc. an alloy such as Nichrome is used, not pure metals.

c. For electric power transmission, copper or aluminium wire is used.

[2023, 2025]

Ans:

1. Copper and aluminium have low resistivity.
2. Therefore, copper and aluminium wires offer low resistance to the flow of current.
3. Hence more amount of current is passed in the circuit with minimum production of heat.
4. As they have large current-carrying capacity, copper or aluminium wires are used for electric power transmission.

d. In practice the unit kWh is used for the measurement of electrical energy, rather than joule.

Ans:

1. The unit of electric power 1W is a very small unit and corresponding energy in joule is also very small.
2. Hence 1000 W or 1 kW is used as a unit to measure electric power, in practice.
3. If 1 kW power is used for 1 hour, it will mean $1\text{ kW} \times 1\text{ hr}$ of electrical energy is used.
4. **1 Unit = 1 kWh = 1 kilowatt hour = $1000\text{ W} \times 3600\text{ s} = 3.6 \times 10^6\text{ Ws} = 3.6 \times 10^6\text{ J}$**

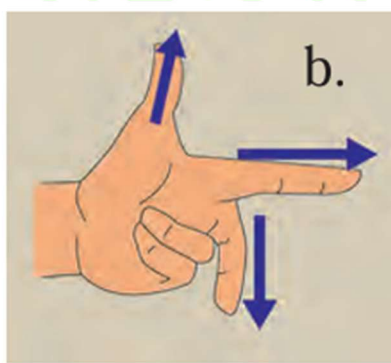
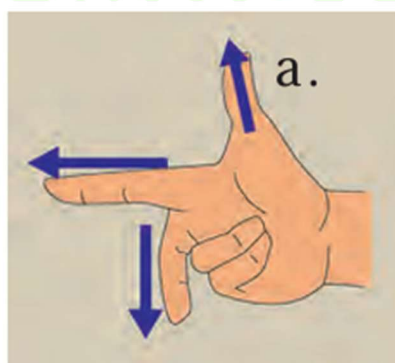
2. Explain the difference: AC generator and DC generator.

AC Generator	DC Generator
1. Ac generator produces alternating current.	1. DC generator produces a steady current.
2. The coil is connected to conducting rings.	2. The coil is connected to split rings

★ **3 Marks Questions:** ★

1. Name the following diagrams and explain the concept behind them.

[2025]



Ans:

a. Fleming's right-hand rule:

Stretch the thumb, the index finger and the middle finger of the right hand in such a way that they will be perpendicular to each other.

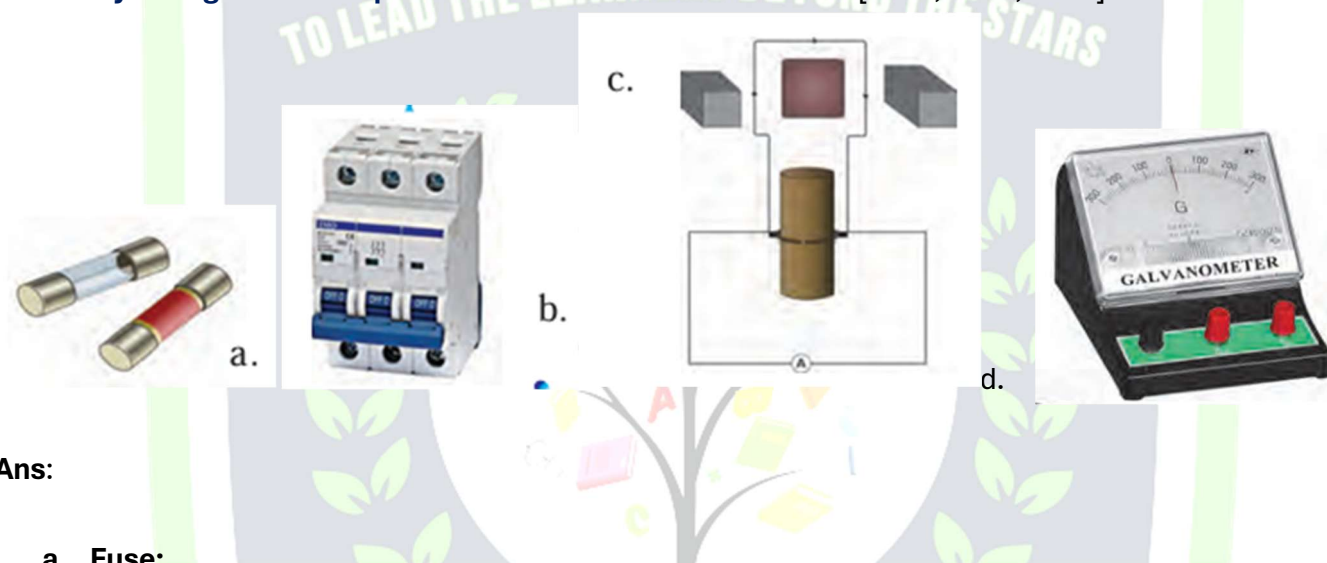
In this position, the thumb indicates the direction of motion of the conductor, the index finger the direction of the magnetic field, and the middle finger shows the direction of the induced current.

b. Fleming's left-hand rule:

According to this rule, the left-hand thumb, index finger, and the middle finger are stretched so as to be perpendicular to each other.

If the index finger is in the direction of the magnetic field, and the middle finger points in the direction of the current, then the direction of the thumb is the direction of the force on the conductor.

2. Identify the figures and explain their use. [2023, 2024, 2025]



Ans:

a. Fuse:

1. A fuse wire is used as a precautionary measure for short circuit.
2. As soon as high current flows in a circuit, the fuse wire melts and breaks the circuit and any mishap is avoided.

b. Miniature circuit breakers (MCB) switches:

1. (MCB) switches are used in homes. When the current in the circuit suddenly increases this switch opens and current stops.
2. Hence it protects against short circuit.

c. DC Generator:

1. A generator converts mechanical energy into electrical energy; it is used to generate electricity.
2. A DC generator produces steady current.

d. Galvanometer:

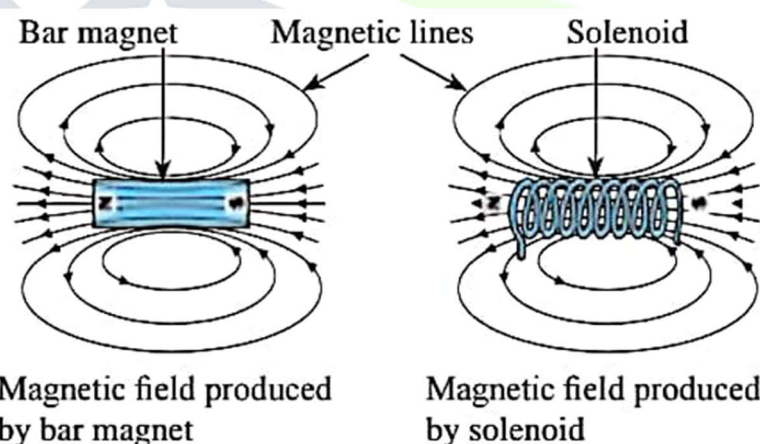
1. Galvanometer is a sensitive device which works on the same principle as that of an electric motor.
2. We can make some electrical measurements with it.

★ 5 Marks Questions: ★

1. What is a solenoid? Compare the magnetic field produced by a solenoid with the magnetic field of a bar magnet. Draw neat figures and name various components. [2024]

Ans:

1. A coil of insulated wire with many turns, wound in the shape of a hollow cylinder is called a solenoid.
2. The properties of magnetic lines of force are similar for the solenoid and the bar magnet.
3. The magnetic lines of force in case of both the solenoid and the bar magnet, are closed curves that originate from the north pole.
4. The magnetic lines of force go to the south pole outside the magnet, and from the south pole to the north pole inside the magnet.
5. The magnetic field inside the solenoid and the bar magnet is uniform as the magnetic lines are parallel.
6. The magnetic field is stronger at the poles than at the centre for both.



II] Extra - Previous Years' Questions: (PYQs)

★ 1 Mark Objectives: ★

1. _____ device is used to produce electricity. [2023, 2025]
(A) Electric motor (B) Galvanometer **(C) Electric generator** (D) Voltmeter
2. _____ is used to make solenoid type of coil in electric bulb. [2025]
(A) Nichrome (B) Copper **(C) Tungsten** (D) Aluminium
3. The unit of electric power is _____. [2024]
(A) Joule (B) Volt **(C) Watt** (D) Coulomb
4. The resistance of wire is $100\ \Omega$. If it is carrying a current of 1 A for 10 seconds, the heat produced will be _____. [2026]
(A) 10 J (B) 10000 J (C) 0.1 J **(D) 1000 J**
5. Name two appliances which work on the principle of magnetic effect of electric current. [2024]

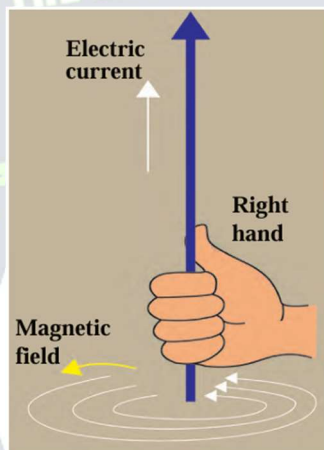
Ans: Fans, mixers, computers etc.

6. Give the unit of intensity of magnetic field. **Ans:** Oersted [2023]

7. **Correlation:** Resistance: ohm:: Potential difference: **volt** [2023]

8. **Match the column:** [2022]

Column A	Column B
Electric Current	a. Ohm b. Ampere c. Volt



9. Identify the law from the figure: **Ans:** Right hand thumb rule [2022]

10. **True or False:** Electric motor converts mechanical energy into electrical energy. **(False)** [2022]

MASTER CLASSES

★ 2 Marks Questions: ★

Q. Give scientific reasons:

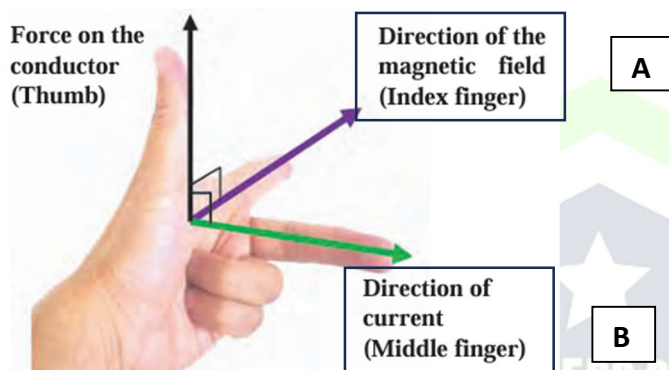
1. A magnetic needle shows decreasing deviation of its angle as distance from a current carrying conductor is increased. [2022]

Ans:

- When current flows through a conductor then magnetic field is produced around it called magnetic effect of electric current.
- The deflection of magnetic needle depends on the strength of the magnetic field produced in the conductor.
- The strength of this magnetic field decreases as the distance between the conductor and the magnetic needle increases.

4. Therefore, a magnetic needle shows decreasing deviation of its angle as distance from a current carrying conductor is increased.

2. Observe the given figure of Fleming's Left Hand Rule and write the labels of "A" and "B". [2022]



★ 3 Marks Questions: ★

1. Observe the following diagram and answer the questions given below. [2022]

a. Identify the device shown in the figure.

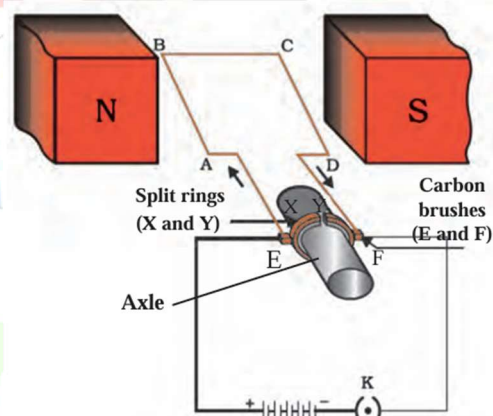
Ans: the device shown in the figure is Electric Motor.

b. On which rule the working of this device based?

Ans: the working of this device is based on Fleming's left-hand rule.

c. Give any two uses of this device.

Ans: Electric motor is used in fans, washing machines, mixers etc.



★ 5 Marks Questions: ★

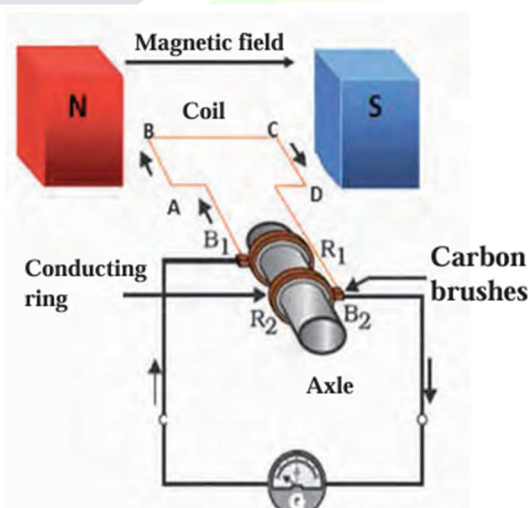
1. Observe the following diagram and answer the questions given below. [2024, 2022]

a. Identify the diagram.

Ans: The diagram is of AC Electric Generator.

b. Write the principle on which the appliance works.

Ans: It works on the principle of Electromagnetic Induction.



c. Write the working of the appliance.

Ans:

1. The coil rotates in clockwise direction and direction of magnetic field is from N to S pole.
2. As per Fleming's right-hand rule, the induced current flows from $A \rightarrow B \rightarrow C \rightarrow D$ and from B2 to B1 in the external circuit.
3. After half rotation the current flows in the direction $D \rightarrow C \rightarrow B \rightarrow A$. Since, B1 is always in contact with branch AB and B2 with CD, this time current flows from B1 to B2 in opposite direction.
4. This repeats in every half rotation producing an alternating current.

d. Write the use of the appliance.

Ans: It is used to produce Alternating electric current.

